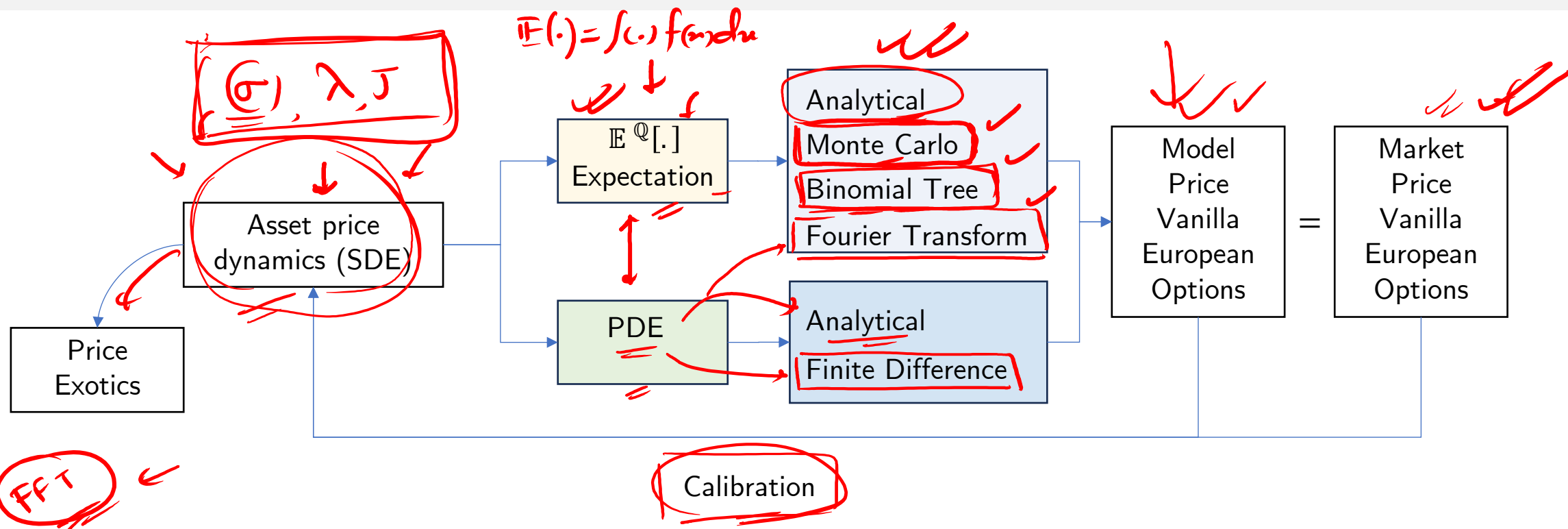


Option Pricing using Fourier Transform

(1. Characteristic functions and Fourier basics)

Pricing approaches

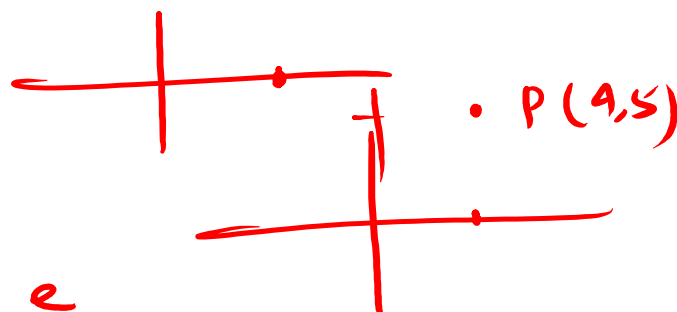


- Analytical solutions are the best (if available) ✓✓
- Monte Carlo shines in high dimension problems (multiple correlated stochastic processes)
- Binomial Tree and Finite difference are good for low dimensional problems (Some Greeks you get free of cost)
- Fourier Transform provides very fast semi-analytical solution and real time calibration capabilities at scale

Complex Number Recap

- Complex numbers is represented as

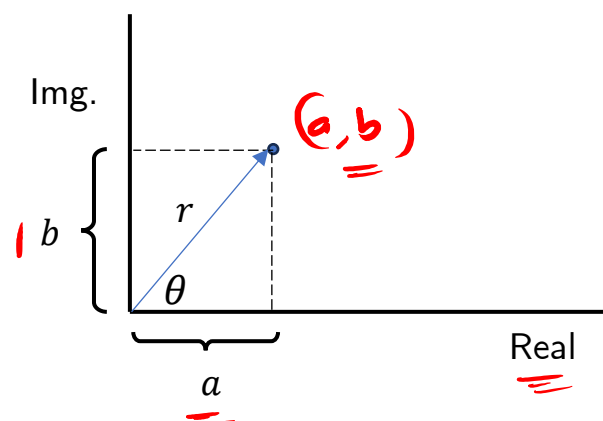
$$x = 5 \quad p = (4, 5)$$



$$z = \underbrace{a}_{\text{Real}} + \underbrace{ib}_{\text{Imaginary}}$$

$$i = \sqrt{-1}$$

$$i^2 = -1$$



- Alternatively, z can be written as $z = re^{i\theta}$ where $r = \sqrt{a^2 + b^2}$ and $\tan \theta = b/a$

- Euler's identity $e^{i\theta} = \cos\theta + i\sin\theta$

Characteristic function and Fourier transform

- The Fourier transform and Inverse Fourier transform of a function called the **Fourier pair** is defined as

Fourier Transform

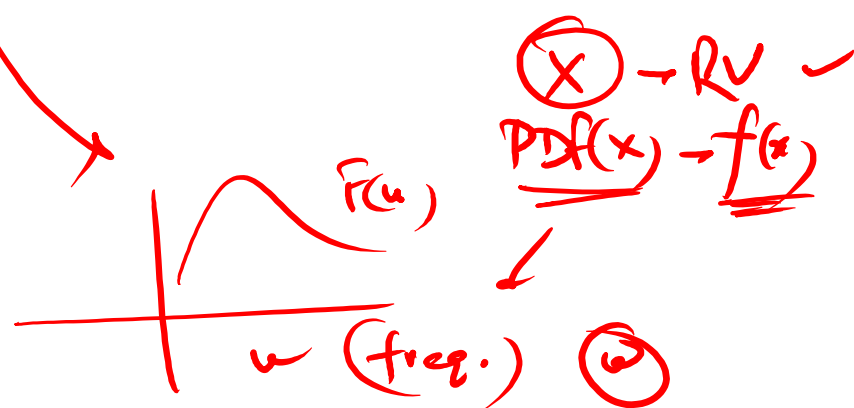
$$\underline{F(u)} = \int_{-\infty}^{\infty} \underline{e^{iux}} \underline{f(x)} dx$$

Inverse Fourier Transform

$$\underline{f(x)} = \left(\frac{1}{2\pi} \right) \int_{-\infty}^{\infty} \underline{e^{-iux}} \underline{F(u)} du$$



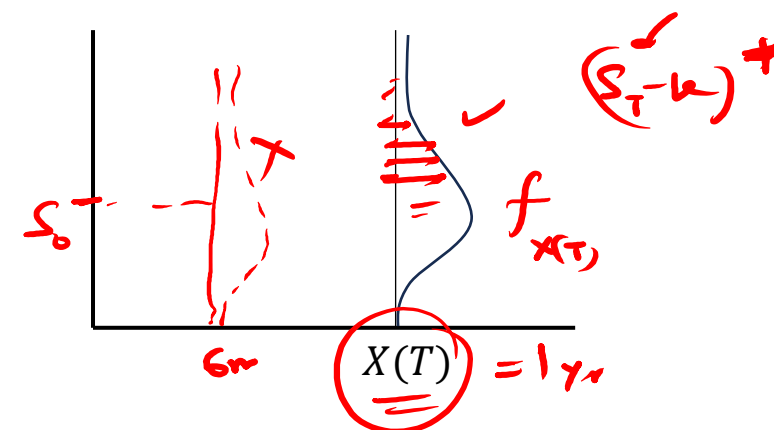
- The characteristic function is the **Fourier transform of the density** function. For European option pricing, we are concerned only with the terminal density $f_{X(T)}(x)$



Handwritten diagram showing the relationship between $\phi_X(u)$ and $f_X(x)$. $\phi_X(u)$ is plotted against u and $f_X(x)$ is plotted against x . A red arrow points from $f_X(x)$ to $\phi_X(u)$. A red circle labeled 'CF' is next to the text 'The characteristic function is the Fourier transform of the density function'.

$$\underbrace{\phi_X(u)}_{\text{Characteristic function}} = \mathbb{E}[e^{iux}] = \int_{-\infty}^{\infty} e^{iux} \underbrace{f_X(x)}_{\text{PDF}} dx$$

$$= \int_{-\infty}^{\infty} e^{iux} \cdot f(x) du$$



Properties of characteristic functions

- Moments can be recovered via differentiation

$$\mathbb{E}[X^k] = \frac{1}{i^k} \frac{\partial^k \phi_X(u)}{\partial u^k} \Big|_{u=0} \quad (\text{Prove it!})$$

- If X and Y are independent, then CF of $X + Y$ will result in product of CFs

$$\phi_{X+Y}(u) = \phi_X(u) \phi_Y(u)$$

- Scaling of a random variable

$$\phi_{aX}(u) = \phi_X(au)$$

Exercise : Show that the characteristic function for a Normal Random $X \sim N(\mu, \sigma^2)$ is given as $\phi_X(u) = e^{iu\mu - \frac{1}{2}\sigma^2 u^2}$

Pricing via Fourier Techniques

- Price of European options depend on the **terminal density** of the stock price. For some cases, the terminal density is available in closed form (Black-Scholes lognormal). But for many other models the density is not available in closed form, however the characteristic function is known. Fourier techniques can price options directly from characteristic function.

$f(x)$ SDF →

Using the convention $\phi_X(u; \tau) = \mathbb{E}_t[e^{iuX_T}]$, $X_T = \ln S_T$, $x = \ln S_t$, $\tau = T - t$

$$\mathbb{E}^Q(e^{-r\tau} (S_T - K)^+) = \int e^{-r\tau} (S_T - K)^+ f(S_T) dS_T$$

✓ Black-Scholes $\phi_X(u) = \exp\left(iu\left[x + \left(r - q - \frac{1}{2}\sigma^2\right)\tau\right] - \frac{1}{2}\sigma^2 u^2 \tau\right)$

✓ Bachelier / Normal model Usually written for price/forward F_T , not log-price: $\phi_F(u) = \exp\left(iuF_t - \frac{1}{2}\sigma_N^2 u^2 \tau\right)$

✓ Merton Jump Diffusion $\phi_X(u) = \exp\left(iu\left[x + \left(r - q - \lambda\kappa - \frac{1}{2}\sigma^2\right)\tau\right] - \frac{1}{2}\sigma^2 u^2 \tau + \lambda\tau\left(e^{ium_J - \frac{1}{2}\delta_J^2 u^2} - 1\right)\right)$ where $\kappa = e^{m_J + \frac{1}{2}\delta_J^2} - 1$

✓ Variance Gamma $\phi_X(u) = \exp(iu[x + (r - q - \omega)\tau]) \left(1 - iu\theta v + \frac{1}{2}\sigma^2 v u^2\right)^{-\tau/v}$ where $\omega = -\frac{1}{v}\ln\left(1 - \theta v - \frac{1}{2}\sigma^2 v\right)$

✓ Heston $\phi_X(u) = \exp(iux + iu(r - q)\tau + C(u, \tau) + D(u, \tau)v_t)$

✓ Bates $\phi_X^{\text{Bates}}(u) = \phi_X^{\text{Heston}}(u; r - q - \lambda\kappa) \times \exp\left(\lambda\tau\left(e^{ium_J - \frac{1}{2}\delta_J^2 u^2} - 1\right)\right)$

Properties of Fourier Transform

In Signal processing, Fourier transform transforms a signal from time domain (t) to frequency domain (ω). Inverse Fourier transforms it back to time domain. For our application, we are using x instead of t and u instead of ω . Here are some the important properties of Fourier transform

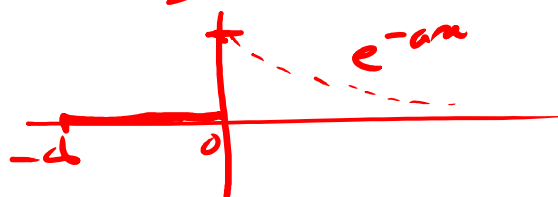
$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

Property	Formula
Linearity ✓	$\mathcal{F}\{af_1(x) + bf_2(x)\} = aF_1(u) + bF_2(u)$
Shift in x ✓	$\mathcal{F}\{f(x - x_0)\} = e^{iux_0} F(u)$
Shift in u	$\mathcal{F}\{f(x)e^{iu_0x}\} = F(u + u_0)$ ✓
Scaling in x	$\mathcal{F}\{f(ax)\} = \frac{1}{ a } F\left(\frac{u}{a}\right)$ ✓
Reversal in x	$\mathcal{F}\{f(-x)\} = F(-u)$ ✓
Differentiation in x	$\mathcal{F}\left\{\frac{d}{dx} f(x)\right\} = -iuF(u)$ ✓

Common Fourier pairs

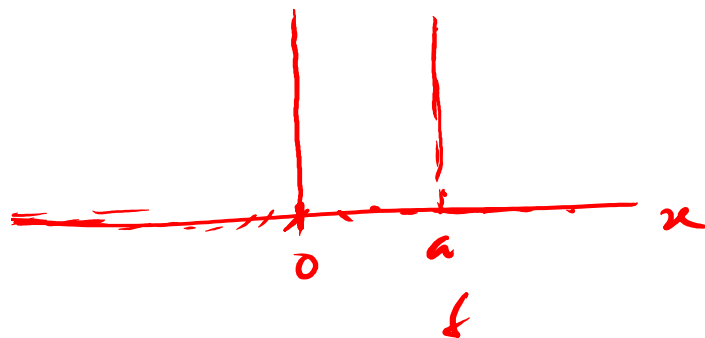
The Fourier pair for some commonly used functions are given below. Note that if the function is spread out in x domain, its Fourier transform will be localized (impulse) in u domain and vice versa.

Function : $f(x)$	Fourier Transform : $F(u)$
$\delta(x)$ ← ←	1
$\delta(x - a)$	e^{iua}
1	$2\pi\delta(u)$ ✓
e^{-iax}	$2\pi\delta(u - a)$ ✓
$\cos(ax)$	$\pi[\delta(u - a) + \delta(u + a)]$ ✓
$\sin(bx)$	$i\pi[\delta(u - b) - \delta(u + b)]$ ✓
$e^{-ax}\mathbf{1}_{x \geq 0}, \quad a > 0$	$\frac{1}{a - iu}$ ✓



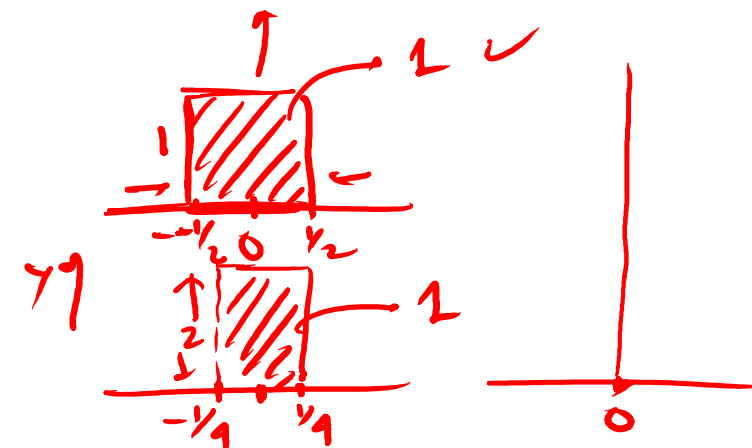
The Dirac Delta function

- Let's get familiar with Fourier transform of some basic functions. To begin with we need to understand the **Dirac Delta** function. It's best understood as infinitely narrow spike with area 1. It's defined as

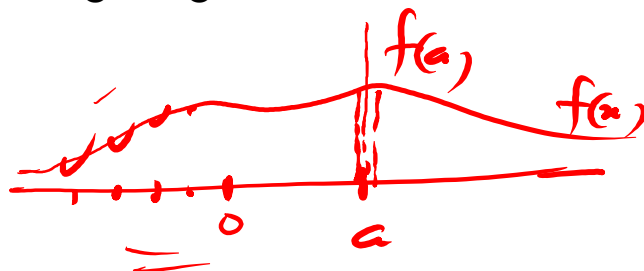


$$\delta(x) = 0, \quad x \neq 0$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$



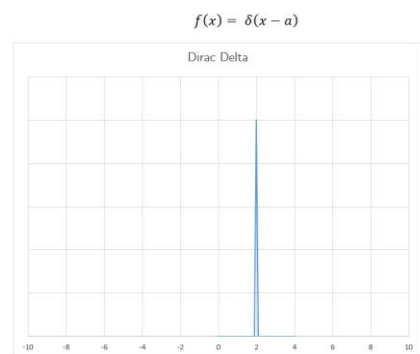
- Shifted delta function $\delta(x - a)$ which will have a spike at $x = a$
 $(x=a)$ $(x \neq a)$
- Integrating a function times the delta function just **picks the value of the function where the spike is located**.



$$\int_{-\infty}^{\infty} \delta(x - a) f(x) dx = f(a)$$

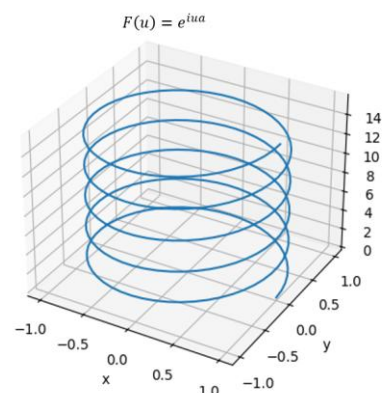
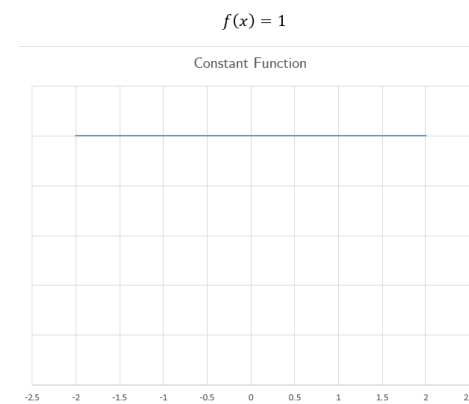
Exercise : Show that the Fourier transform of delta function is 1

Fourier Transform Visuals



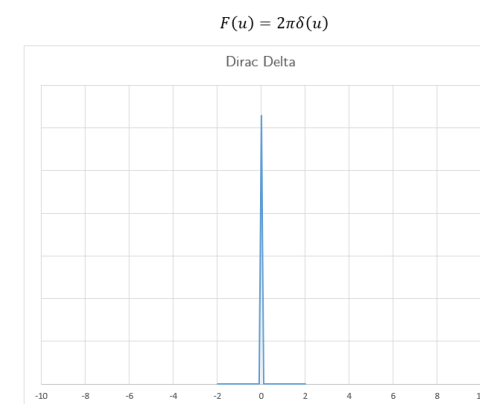
$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} F(u) du$$

 u 

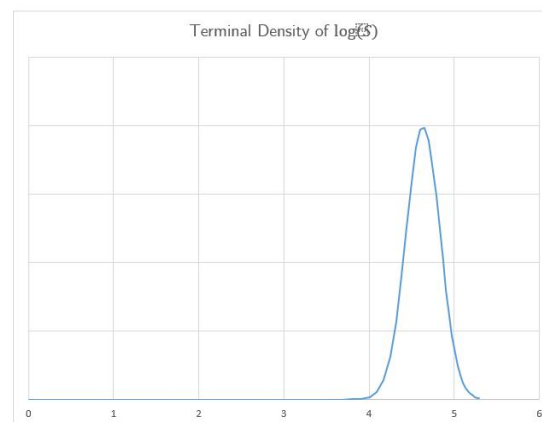
$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} F(u) du$$



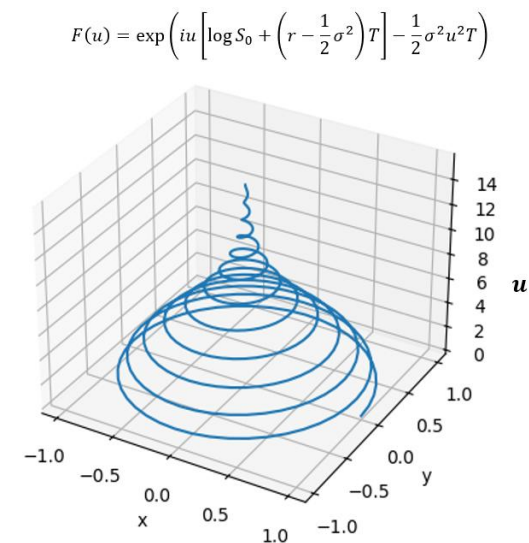
$X = \log(S_T)$

$$X \sim N\left(\log S_0 + \left(r - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right)$$



$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} F(u) du$$

 u